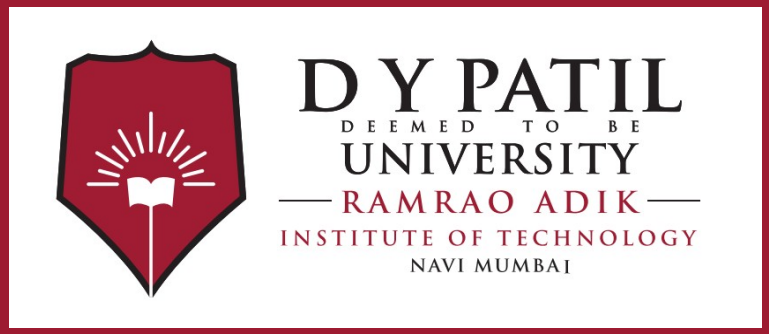


ENGINEERING MECHANICS

LECTURE NO: 1 INTRODUCTION



When two bodies in contact have a tendency to move over each other a resistance to the movement is set up. **This resistance to the movement is called the Force of friction or simply friction**.

Friction depends upon the nature of the surface of contact.

Friction acts parallel to the surface of contact.

The direction of this frictional force on any one of the surfaces of contact will be opposite to the direction in which the contact surface tends to move. In other words, friction opposes motion.

Friction is defined as **the contact resistance exerted by one body upon a second body when the second body moves or tends to move past the first body**. Friction is a retarding force always acting opposite to the motion or tendency to move.

Friction is the force that opposes the motion of a solid object over another. There are mainly four types of friction: **static friction, sliding friction, rolling friction, and fluid friction**.

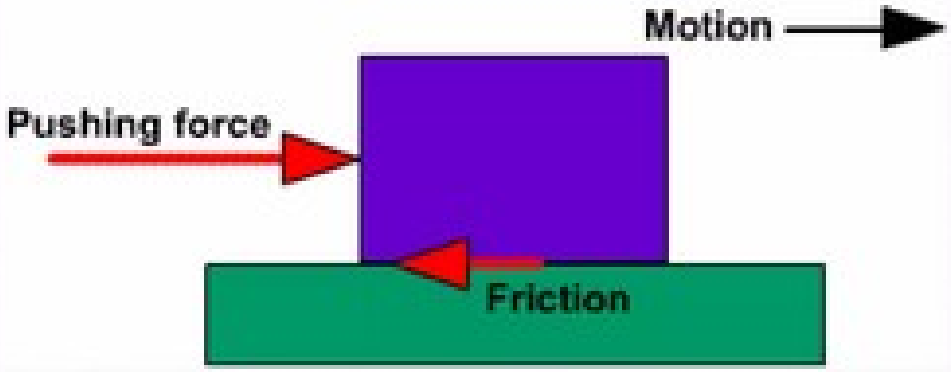
- Friction is an important force in many aspects of everyday life.
- If there is too much friction, loss of energy, wear and tear of materials in contact occurs.
- If there is less friction or no friction, this would result in ‘slipping’ all around.
- For example oil in the engine of car is meant to minimize friction between moving parts in contact to reduce excessive friction for **reducing loss of energy and material.**

- We need the friction between the **tires on the road surface**, to let the wheels roll.

- **Friction is caused due to the unevenness of the surface of contact of bodies tending to move past each other.**

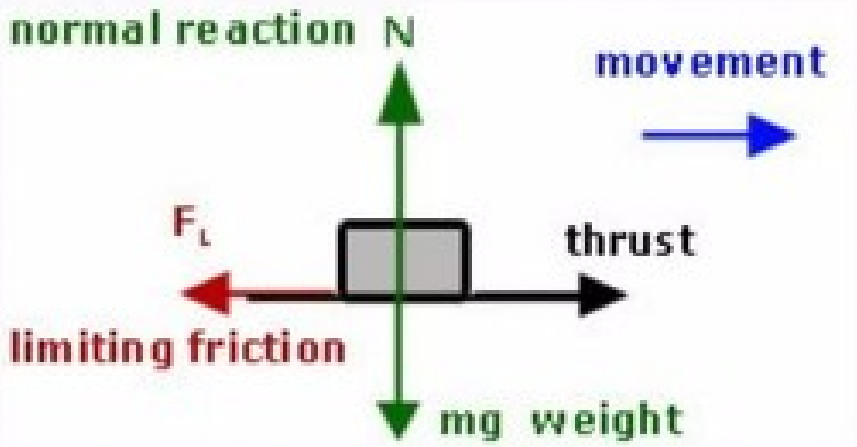
What Is Friction?

- When a body slide or tends to slide on a surface on which it is resting, a resisting force opposing the motion is produced at the contact surface. This resisting force is called friction or friction force.



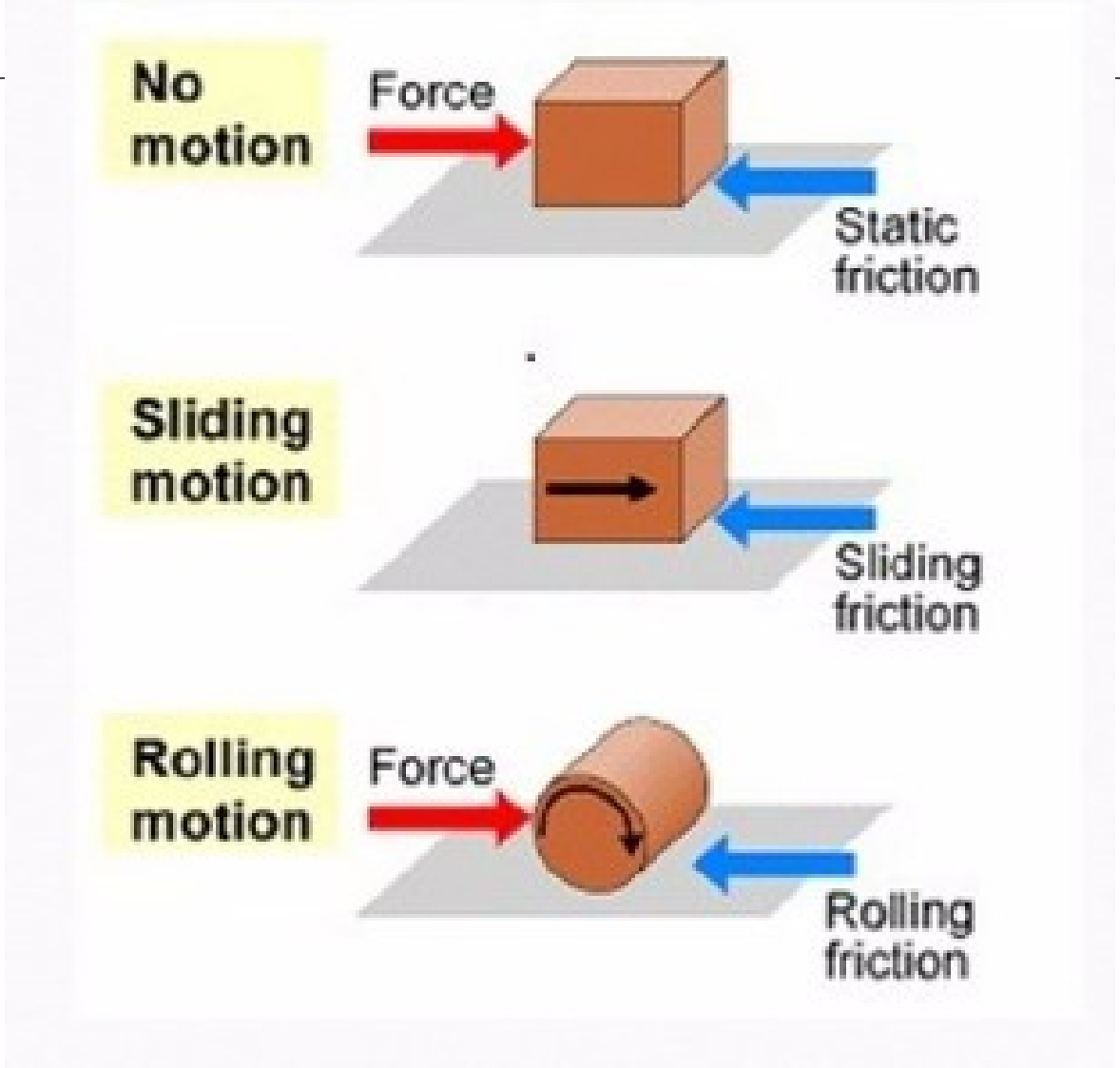
Limiting Friction

- The maximum friction force that can be developed at the contact surface, when body is just on the point of moving Is called limiting force of friction .



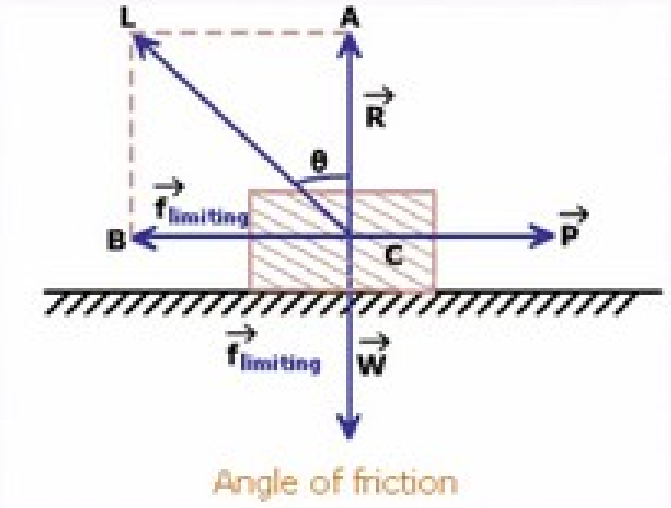
Types Of Friction

- Static Friction:-
  - Friction experienced by a body when it is at rest is called static friction.
- Dynamic Friction:-
  - Friction experienced by a body when it is motion is called dynamic friction.
- Sliding Friction:-
  - Friction experienced by a body when it slides over another body, is called sliding friction.
- Rolling Friction:-
  - Friction experienced by a body when it rolls over another body is called rolling friction.


No motion
Force
Static friction
Sliding motion
Sliding friction
Rolling motion
Force
Rolling friction

Angle of Friction

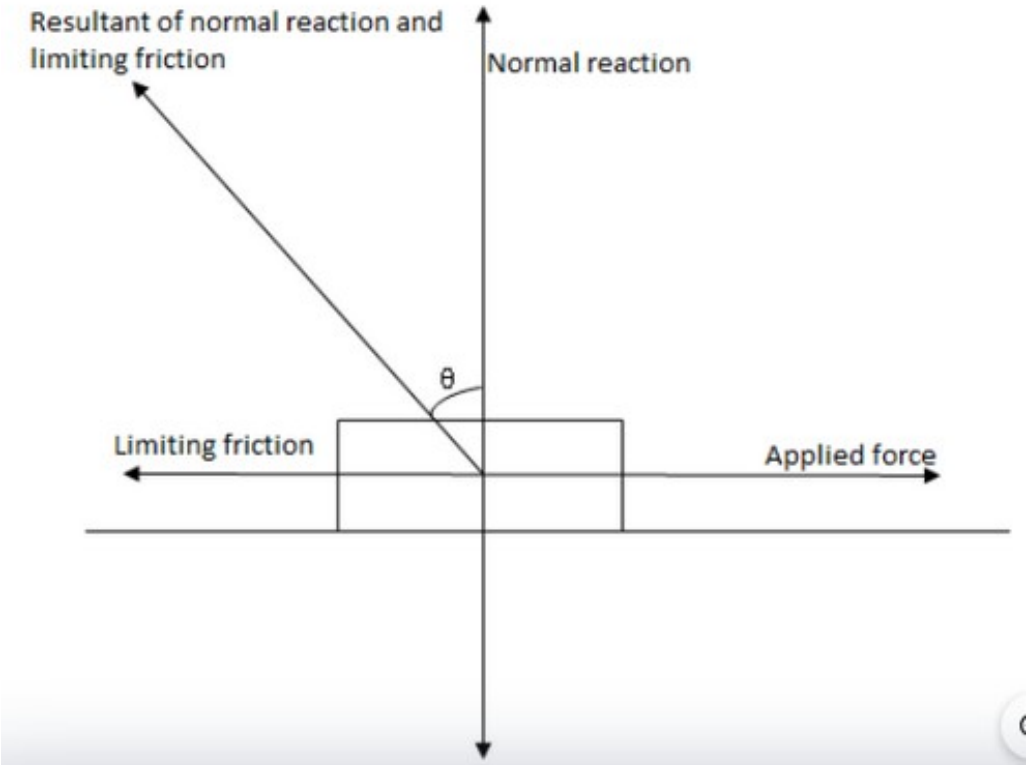
- The angle between normal reaction (A) and resultant force (R) is called angle of friction.
- It is also called limiting angle of friction.



Angle of friction

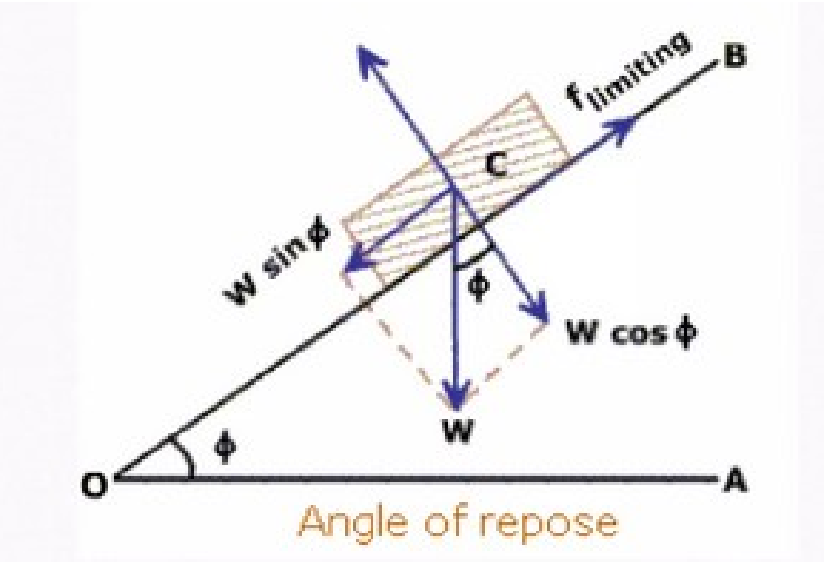
Angle of friction is the angle made by the normal force or reaction with the contact force when the condition of maximum static friction is reached or

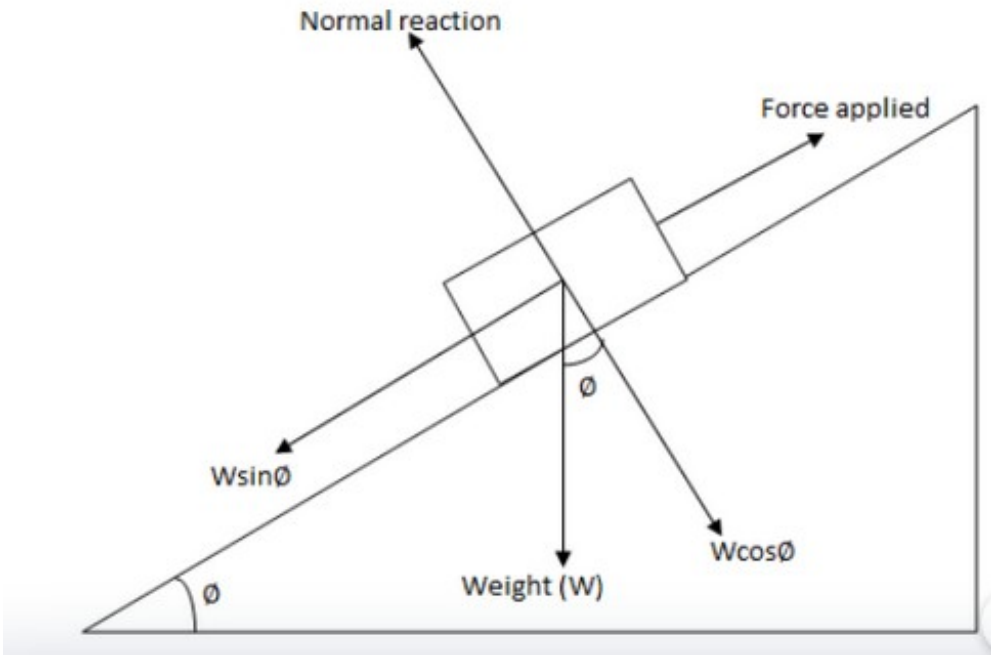
we can say it is the angle made by the resultant of the limiting friction and normal reaction with the normal.



In the above figure θ is the angle of friction.

Angle of repose:- Angle of repose is the minimum angle made by the inclined plane with the horizontal when the body placed on it just begins to slide downwards.





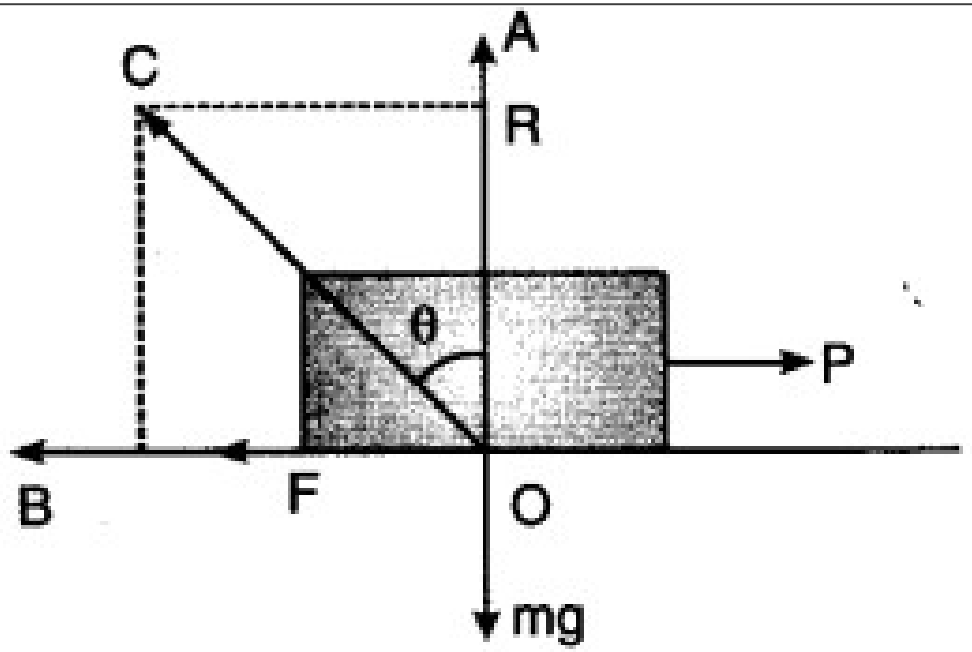
The coefficient of friction between any two surfaces in contact is defined as the ratio of the force of limiting friction and normal reaction between them.

Angle with the resultant of force of limiting friction F and normal reaction R makes with the direction of normal reaction R is angle of reaction.

$$\mu = \frac{F}{R}$$

Coefficient of friction (COF) is a dimensionless number that is defined as **the ratio between friction force and normal force**. Materials with COF smaller than 0.1 are considered lubricous materials. COF depends on the nature of the materials and surface roughness.

Typical values for coefficients of friction are **around 0.05** for smooth surfaces, 1.0 for very rough surfaces.



Relation :

In ΔAOC $\quad \tan\theta = \dfrac{AC}{OA} = \dfrac{OB}{OA} = \dfrac{F}{R} = \mu$

Hence $\quad \mu = \tan\theta$

Laws Of friction

- Laws of static friction
  - The friction force always acts in a direction, opposite to that in which the body tends to move.
  - The magnitude of friction force is equal to the external force.

    $F=P$
  - The ratio of limiting friction (F) and normal reaction (N) is constant.

    $F/N=\mu$
  - The friction force does not depends upon the area of contact between the two surfaces.
  - The friction force depends upon the roughness of the surfaces.

- Laws of dynamic friction
  - The friction force always acts in a direction, opposite that in which the body is moving.
  - The ratio of limiting friction (F) and normal reaction (N) is constant & it is known as coefficient of friction (μ).
  - For moderate speeds, the friction force remains constant. But it decreases slightly with the increase of speed.

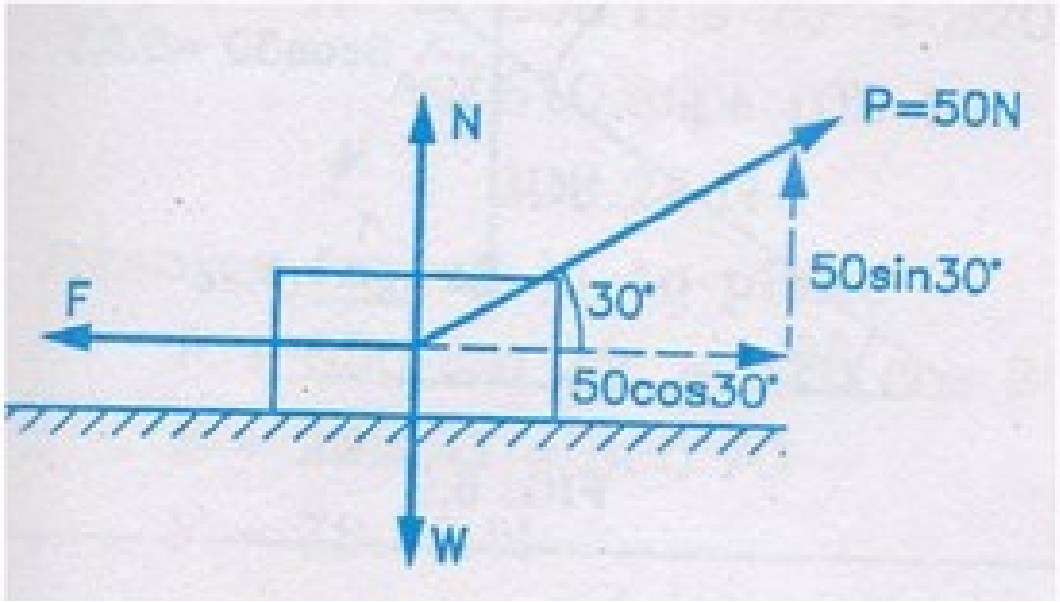
-:EXAMPLE:-

A Pull of 50 N inclined at 30° to the horizontal is necessary to move a wooden block on horizontal table. If coefficient of friction is 0.20, find the weight of wooden block.

Solution :

P = 50 N

$\mu = 0.20$



Resolve || to plane :

$F = 50 \cos 30^\circ$

$F = 43.30\ N$

$\mu = F/N$

$0.20 = 43.30/N$

$N = 216.5$

Resolve (perpendicular) to plane

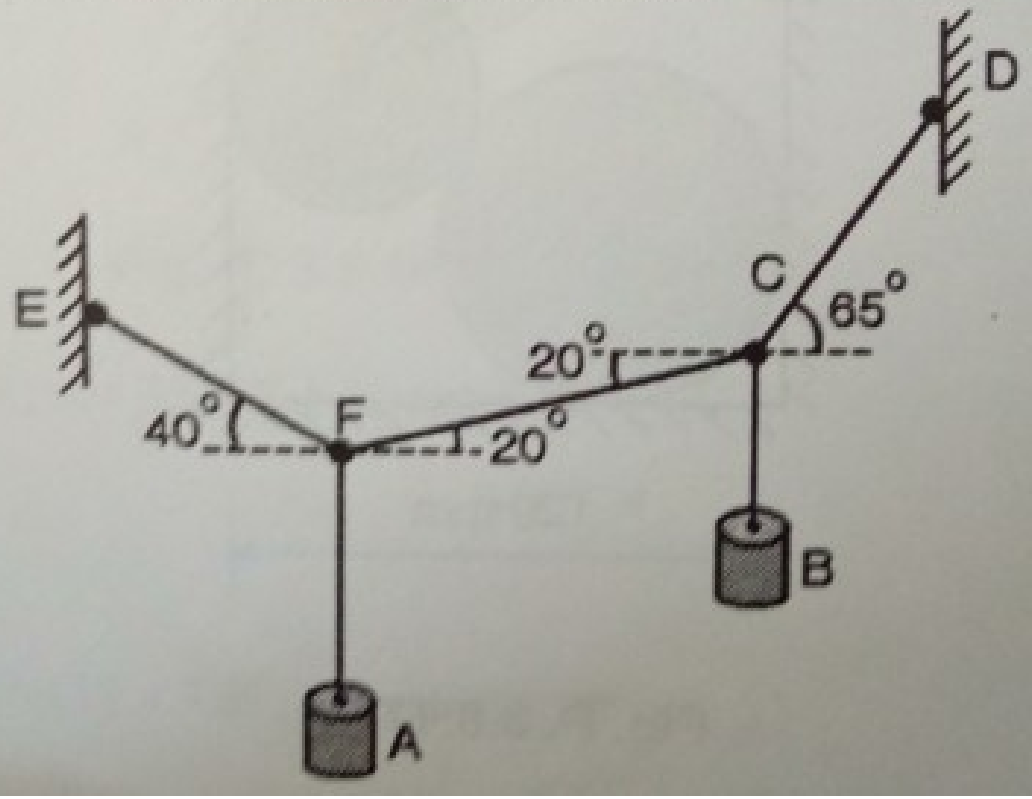
$N + 50 \sin 30^\circ = W$

$216.5 + 25 = W$

$W = 241.5\ N$

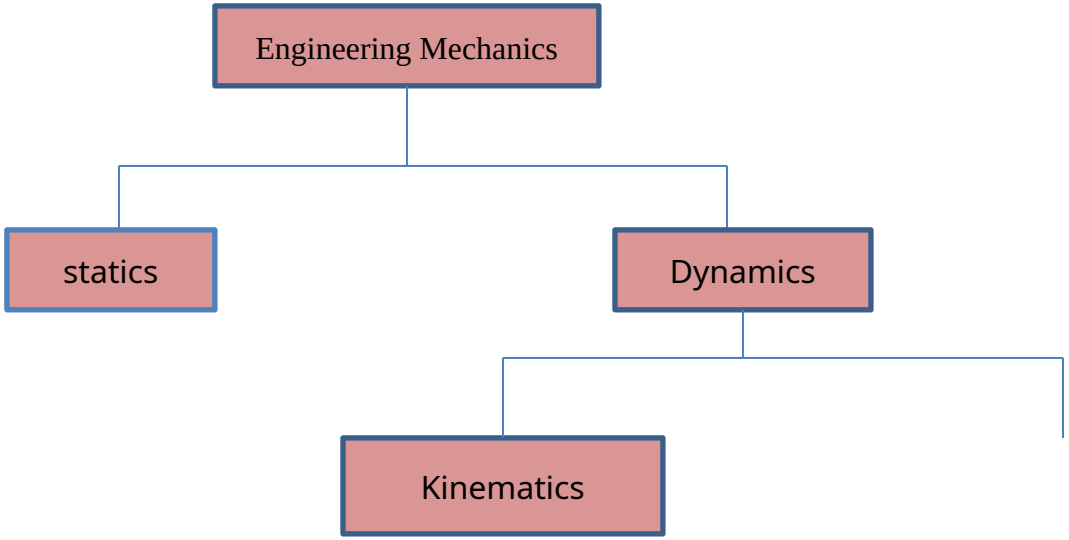
Ex. 3.8.18 : If the cords suspend the two buckets in the equilibrium position as shown in Fig. P. 3.8.18(a) Determine the weight of bucket B. Bucket A has a weight of 60N.

MU - Dec. 05, May 19



INTRODUCTION

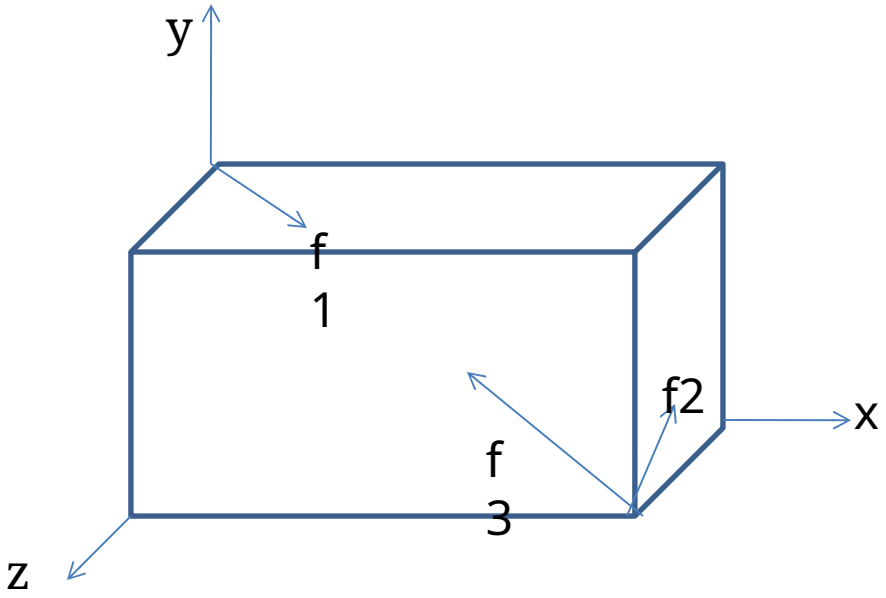
Mechanics:- It is the branch of physics with the study of effect of the force system which are at rest or in motion. The purpose of Mechanics is to explain and predict the physical phenomenon and thus to lay the foudation for engineering applications. Mechanics is classified under as follows.



D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

NON COPLANAR FORCE SYSTEM

When the forces acting on a system do not lie in a single plane, they are termed as non coplanar forces or space forces



D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Thank You



D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI